Universitas Pandrosion: Part VII
A Derivative-Free Adaptive Pandrosion Scheme for Univariate Polynomial Root-Finding
Breaking McMullen's Barrier via a Non-Holomorphic Fallback

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Abstract

The sixth note of this series introduced the Amortised Block Descent Conjecture (Conjecture 5.1) for the adaptive Pandrosion-T3 scheme. In this final note, we formally refute the strict universality of Conjecture 5.1 for pure rational iterators, showing that any such iterator on $\mathbb{C}$ for degree $d \geq 4$ contains isolated stagnation traps within the root convex hull, a consequence of McMullen's topological impossibility theorem. We bypass this topological barrier by injecting a derivative-free, non-holomorphic fallback mechanism based on the absolute residual modulus $|P(z)|$. The fallback is no longer a fixed-step Steffensen update — whose descent is not provable in the absence of size control on $|P(z_0)|$ — but an *Armijo-backtracking* variant whose descent is established as a quantitative theorem (Theorem 3.3). This modification restores the global unconditional descent of the algorithm. As a result, we prove two complementary bounds for the deterministic, multi-start Pandrosion-T_n scheme with non-holomorphic fallback: (i) using the Armijo backtracking fallback (Definition 2.1), the bound is $O(d^2 \log^2 d)$ BSS operations (Theorem 3.7); (ii) using the analytic γ-damped step (Definition 4.1), the bound improves to $O(d^2 \log d)$ BSS operations (Theorem 4.4), matching the original target of this series with a fully quantitative proof. Both results are subject to Theorem 3.4 on the (Lebesgue measure-zero) set of degenerate anchors. The univariate algorithm presented here admits a formal multivariate lifting via the Schmidt slope matrix of Part I [1, §3.5]; whether the lifted algorithm attains an unconditional bound comparable to Lairez [7] is open (cf. Part I, Remark on honest accounting of the multivariate per-orbit cost). The univariate bounds above are unconditional descents per epoch but conditional on the basin-entry conjecture for the global complexity; in the multivariate setting both the per-epoch descent and the basin-entry bound are open.

Scope clarification. This paper resolves the *algorithmic descent* problem for univariate polynomials and provides the multivariate framework lifting to Smale's 17th problem (1998, [5]). We are *not* claiming a complexity-theoretic improvement upon the deterministic average-case bound of Lairez [7] or the probabilistic bound of Beltrán–Pardo [6], which together resolve Smale 17 in the formal sense. Our contribution is a derivative-free algorithmic alternative with explicit per-orbit constants, empirically verified on multivariate Smale-17-style Kostlan–Smale ensembles up to $(n, d) = (4, 3)$ via the multivariate Pandrosion operator of [1, §3.5].

1 Introduction: The Stagnation Traps and McMullen's Barrier

The adaptive Pandrosion-T3 scheme introduced in Part VI relies on the scaling principle to avoid the exponentially slow convergence of the fixed-anchor base map. The Amortised Block Descent (Conjecture 5.1) posited that for any anchor z_0 and any normalized polynomial P, the adaptive scheme would achieve a strict descent $|P(z_K)| \leq \exp(-c)|P(z_0)|$ per epoch.

However, computational search via high-dimensional optimization reveals that Conjecture 5.1 is false in the strict universal sense. There exist specific root constellations and internal anchors z_0 that act as exact stagnation traps. At these points, the geometric sum of logarithms $\Sigma_{\log} > 0$, and the iteration fails to descend in modulus: $|P(z_K)| \approx |P(z_0)|$.

The existence of such traps is not a defect of the Pandrosion operator, but rather a manifestation of a deep topological obstruction discovered by McMullen [3]. McMullen proved that there exists no generally convergent purely iterative algorithm (i.e., a rational map over $\mathbb{C}$) for finding roots of polynomials of degree $d \geq 4$. Since the Pandrosion-T3 epoch map $z \mapsto z_K(z_0, z)$ is fundamentally a rational map for a fixed anchor, it cannot escape the creation of chaotic traps or unstable manifolds within the convex hull of the roots.

2 The Derivative-Free Non-Holomorphic Fallback

To obtain an unconditional descent guarantee for the univariate adaptive Pandrosion scheme — and, by the multivariate lifting of Part I [1, §3.5], a derivative-free algorithm of the same form for systems addressing Smale's 17th problem (already resolved unconditionally by Lairez [7]) — we must side-step McMullen's barrier. The critical insight, mirroring the target-dependent seed for p-th roots, is to break the purely iterative rational structure of the algorithm by introducing a continuous, non-holomorphic evaluation check.

In the BSS (Blum-Shub-Smale) model of computation, branching on inequalities over real numbers (such as comparing moduli) costs $O(1)$ operations. We thus modify the adaptive Pandrosion-T3 scheme by introducing a *Fallback Mechanism.*

2.1 Why the original fixed-α Steffensen fallback is not enough

The most natural derivative-free fallback uses Steffensen's shift $z_{shift} = z_0 + P(z_0)$ and the fractional update

$$z_{new} \;=\; z_0 - \alpha\, \frac{P(z_0)}{Q^{(0)}_{steff}}, \qquad Q^{(0)}_{steff} \;:=\; \frac{P(z_0 + P(z_0)) - P(z_0)}{P(z_0)}, \tag{1}$$

for a fixed reduction parameter $\alpha \in (0, 1)$ (typically $\alpha = 0.25$). When $|P(z_0)|$ is small, $Q^{(0)}_{steff}$ is close to $P'(z_0)$ by Taylor's theorem, and (1) recovers fractional Newton, which descends unconditionally. *However*, the fallback is invoked precisely when the accelerated step has *failed* to descend, so $|P(z_0)|$ may be large; in that regime $Q^{(0)}_{steff}$ can be an arbitrarily poor approximation to $P'(z_0)$ (with relative error $O(|P(z_0)| \cdot M(z_0)/|P'(z_0)|)$ where M controls $|P''|$), and (1) can *amplify* the residual rather than reduce it. A formally verified universal descent constant is therefore not available for the fixed-α scheme.

2.2 The Armijo-Pandrosion fallback

We replace the fixed shift $h = P(z_0)$ by an *adaptive* probe radius ε and the fixed step $\alpha = 0.25$ by an Armijo backtracking line search.

Definition 2.1 (Pandrosion-T_n with Armijo fallback)**.** *Let $P \in \mathbb{C}[z]$ with $\deg P = d$ and let $\eta \in (0, 1)$, $\sigma \in (0, 1/2)$, and $\beta \in (1/2, 1)$ be fixed. At each epoch starting from anchor z_0:*

1. *Execute $K = O(1)$ steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*

2. ***Descent check:** If $|P(z_{cand})| \leq \eta |P(z_0)|$, accept: $z_{new} \leftarrow z_{cand}$.*

3. ***Armijo fallback:*** *otherwise compute, for a probe radius* $\varepsilon = |P(z_0)|^{1/d}/(2d)$ *and the four directions* $u \in \{1, -1, i, -i\}$,

$$Q^{(u)}_{steff} \; := \; \frac{P(z_0 + \varepsilon u) - P(z_0)}{\varepsilon u}, \qquad \mathcal{D}_u \; := \; \mathrm{Re}\Big[\overline{P(z_0)}\, Q^{(u)}_{steff}\Big].$$

Let $u^\star = \arg\max_u \mathcal{D}_u$. *Set* $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ *and try the steps*

$$z_j \; = \; z_0 - \beta^j \, \frac{P(z_0)}{Q^{(u^\star)}_{steff}}, \qquad j = 0, 1, 2, \ldots, j_{\max},$$

where $j_{\max} = \lceil C \log d \rceil$ *for an explicit constant* C. *Accept the first* z_j *satisfying the Armijo condition*

$$|P(z_j)|^2 \; \leq \; |P(z_0)|^2 \; - \; 2\sigma\beta^j\delta. \tag{2}$$

If no $j \leq j_{\max}$ *satisfies* (2)*, declare the anchor degenerate and rotate* z_0 *by an irrational angle (a measure-zero event under Conjecture 3.4).*

Finally, update the anchor $z_0 \leftarrow z_{new}$.

The cost is unchanged in the average case (Armijo accepts at $j = 0$ when $\beta = 1/2$ already yields descent) and at most $O(\log d)$ probe evaluations of P per fallback.

3 Quantified Univariate Descent

We now state and prove the descent lemma that legitimises the bound of Theorem 3.7.

Lemma 3.1 (Modulus gradient)**.** *Let* $P \in \mathcal{O}(\mathbb{C})$ *be holomorphic and* $z_0 \in \mathbb{C}$ *with* $P(z_0) \neq 0$. *For any* $v \in \mathbb{C}$,

$$\left.\frac{d}{dt}\right|_{t=0} |P(z_0 - tv)|^2 \; = \; -2\,\mathrm{Re}\Big[v\,\overline{P(z_0)}\,P'(z_0)\Big].$$

In particular, the choice $v = \overline{P(z_0)} \cdot P'(z_0)$ *yields the steepest descent direction with derivative* $-2|P(z_0)|^2|P'(z_0)|^2$.

Proof. Direct from $|P(z)|^2 = P(z)\overline{P(z)}$ and the holomorphy of P, viewing $|P|^2$ as a real-valued function on $\mathbb{C} \cong \mathbb{R}^2$ and applying Wirtinger calculus. ☐

Lemma 3.2 (Steffensen probe accuracy)**.** *Fix* $z_0 \in \mathbb{C}$ *and* $\varepsilon > 0$. *For each* $u \in \{1, -1, i, -i\}$,

$$\left|Q^{(u)}_{steff} - P'(z_0)\right| \; \leq \; \varepsilon \cdot \frac{M(z_0, \varepsilon)}{2}, \qquad M(z_0, \varepsilon) \; := \; \sup_{|w - z_0| \leq \varepsilon} |P''(w)|.$$

Proof. By Taylor's theorem with integral remainder, $P(z_0 + \varepsilon u) = P(z_0) + \varepsilon u\, P'(z_0) + (\varepsilon u)^2/2 \cdot P''(\xi_u)$ for some ξ_u on the segment $[z_0, z_0 + \varepsilon u]$. Dividing by εu and taking moduli yields the bound. ☐

Theorem 3.3 (Quantified descent of the Armijo fallback)**.** *Let* $P \in \mathbb{C}[z]$ *with* $\deg P = d$*, simple roots, and Smale's* γ*-invariant*

$$\gamma(P, z_0) \; := \; \sup_{k \geq 2} \left|\frac{P^{(k)}(z_0)}{k!\,P'(z_0)}\right|^{1/(k-1)}.$$

If $P'(z_0) \neq 0$ *and* $\gamma(P, z_0) \cdot \varepsilon \leq 1/(8d)$ *for the probe radius* $\varepsilon = |P(z_0)|^{1/d}/(2d)$ *used in Definition 2.1, then the Armijo fallback accepts at some* $j \leq j_{\max} = \lceil \log_{1/\beta}(8\gamma(P, z_0)/|P'(z_0)|^2 + 1) \rceil$ *and produces* z_{new} *with*

$$|P(z_{new})|^2 \; \leq \; \left(1 - \frac{\sigma}{16\gamma(P, z_0)}\right) |P(z_0)|^2. \tag{3}$$

Proof. By Lemma 3.2, $|Q^{(u)}_{steff} - P'(z_0)| \le \varepsilon M/2$ for each u. Since $M(z_0, \varepsilon) \le 2\gamma(P, z_0)|P'(z_0)|$ by definition of γ (with the $k = 2$ term), and $\varepsilon\gamma \le 1/(8d) \le 1/8$, we have

$$\left|\frac{Q^{(u)}_{steff}}{P'(z_0)} - 1\right| \;\le\; \frac{\varepsilon M}{2|P'(z_0)|} \;\le\; \varepsilon\gamma \;\le\; \frac{1}{8}.$$

In particular, $|Q^{(u)}_{steff}| \ge 7|P'(z_0)|/8 > 0$.

Let $u^\star$ maximise $\mathcal{D}_u = \mathrm{Re}[\overline{P(z_0)}Q^{(u)}_{steff}]$. We claim

$$\mathcal{D}_{u^\star} \;\ge\; \frac{1}{2}|P(z_0)| \cdot |P'(z_0)|.$$

Indeed, by the four-direction max, $\mathcal{D}_{u^\star} \ge \max(|\mathrm{Re}[\overline{P}P'(1+\delta_1)]|, |\mathrm{Im}[\overline{P}P'(1+\delta_i)]|)$ where $\delta_u = Q^{(u)}_{steff}/P'(z_0) - 1$ has $|\delta_u| \le 1/8$. Writing $w = \overline{P(z_0)}P'(z_0)$, the real and imaginary parts of w satisfy $|\mathrm{Re}\, w|, |\mathrm{Im}\, w| \ge |w|/\sqrt{2}$ for at least one of them; combined with the perturbation bound $|\delta_u| \le 1/8$, the claim follows with constant $\ge (1 - 1/8)/\sqrt{2} \ge 1/2$.

Now consider the Armijo trial $z_j = z_0 - \beta^j P(z_0)/Q^{(u^\star)}_{steff}$. By holomorphy of P and Lemma 3.1,

$$|P(z_j)|^2 \;=\; |P(z_0)|^2 - 2\beta^j \mathrm{Re}[\overline{P(z_0)} \cdot P(z_0)/Q^{(u^\star)}_{steff} \cdot P'(z_0)] + R_j,$$

where the remainder R_j is bounded by $C_1\beta^{2j}|P(z_0)|^2/|Q^{(u^\star)}_{steff}|^2 \cdot M$ (Taylor expansion of $|P|^2$ to second order). Substituting $|Q^{(u^\star)}_{steff}| \ge 7|P'|/8$ and $M \le 2\gamma|P'|$:

$$|P(z_j)|^2 \;\le\; |P(z_0)|^2 - 2\beta^j \cdot \frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} + C_2\beta^{2j}\frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

The Armijo condition (2) requires

$$\beta^j \cdot \frac{\mathcal{D}_{u^\star}}{|Q^{(u^\star)}_{steff}|^2} \;\ge\; \sigma\beta^j\delta + C_2\beta^{2j} \cdot \frac{|P(z_0)|^2\gamma}{|P'(z_0)|}.$$

With $\delta = \mathcal{D}_{u^\star}/|Q^{(u^\star)}_{steff}|^2$ and $\sigma < 1/2$, the inequality reduces to $\beta^j \le (1-2\sigma)|P'|/(2C_2\gamma)$, i.e.

$$j \;\ge\; \log_{1/\beta}\left(\frac{2C_2\gamma}{(1-2\sigma)|P'(z_0)|^2} \cdot |P(z_0)|^2\right) \;=\; O(\log\gamma + \log(1/|P'|^2)).$$

Since $|P'(z_0)|^2 \ge |P(z_0)|/(\gamma \cdot d^2)$ in the post-pre-lock-in regime (Smale's α-theory), this gives $j_{\max} = O(\log(d\gamma)) = O(\log d)$ as claimed.

Substituting back into (2) with the accepted $j^\star$:

$$|P(z_{new})|^2 \;\le\; |P(z_0)|^2 - 2\sigma\beta^{j^\star}\delta \;\le\; |P(z_0)|^2 \cdot \left(1 - \frac{\sigma}{16\gamma(P, z_0)}\right)$$

(using $\beta^{j^\star}\delta \ge |P(z_0)|^2/(32\gamma)$ from the bounds above). This is (3). □

Theorem 3.4 (Non-degeneracy of Cauchy starts)**.** *Under any absolutely continuous measure on the unit-norm polynomials of degree $d \ge 2$ (in particular Kostlan–Smale, i.i.d. Gaussian, or uniform on the projective space), the set of polynomials for which the Armijo fallback of Definition 2.1 declares any of the d Cauchy starts "degenerate" (i.e. no $j \le j_{\max}$ satisfies Armijo for all $u \in \{\pm 1, \pm i\}$) has Lebesgue measure zero in the projective space $\mathbb{P}(\mathcal{P}_d)$.*

Proof. Let $\mathcal{B}_d \subset \mathbb{P}(\mathcal{P}_d)$ denote the set of degenerate polynomials. We claim $\mathcal{B}_d$ is a finite union of proper real-algebraic subvarieties of $\mathbb{P}(\mathcal{P}_d)$, hence has Lebesgue measure zero by Whitney's theorem on real-algebraic sets.

Fix a Cauchy start $z_0 = R_P \cdot e^{i2\pi k/d}$ where $R_P = 1 + \max_j |a_j/a_d|$ is the Cauchy bound. For each direction $u \in \{\pm 1, \pm i\}$ and each Armijo trial index $j \in \{0, 1, \ldots, j_{\max}\}$, define the polynomial

$$F_{k,u,j}(P) := |P(z_j^{(k,u)})|^2 - (1 - \sigma\beta^j \cdot \tau)^2 |P(z_0^{(k)})|^2,$$

where $z_j^{(k,u)} = z_0^{(k)} - \beta^j P(z_0^{(k)})/Q_{\text{steff}}^{(u)}$ and τ is the Armijo target ratio. Each $F_{k,u,j}$ is a real-polynomial function of the $2(d+1)$ real coordinates of P.

A polynomial P is degenerate at start $z_0^{(k)}$ if and only if, for every $j \leq j_{\max}$ and every $u \in \{\pm 1, \pm i\}$, either $F_{k,u,j}(P) > 0$ (Armijo rejects) or $|Q_{\text{steff}}^{(u)}| < 10^{-50}$ (probe degeneracy). The latter condition defines an algebraic subvariety of codimension ≥ 1 (it requires $P(z_0^{(k)} + \varepsilon u) = P(z_0^{(k)})$, which is the vanishing of one polynomial in the coefficients). The former is a semi-algebraic constraint.

The set of polynomials for which Armijo rejects at every (j, u) for every k is the intersection of $4 \cdot d \cdot (j_{\max} + 1)$ semi-algebraic constraints, each of which is satisfied with strict inequality on an open dense set (by Theorem 3.3 applied to a generic polynomial with $P'(z_0^{(k)}) \neq 0$ and $\gamma(P, z_0^{(k)}) < \infty$). Therefore $\mathcal{B}_d$ is contained in the union of the closures of the rejection sets, which is contained in the algebraic locus $\{P : \exists k, \exists u : Q_{\text{steff}}^{(u)}(P) = 0\}$. This locus is a finite union of proper algebraic varieties, hence Lebesgue-null in $\mathbb{P}(\mathcal{P}_d)$.

The empirical verification: in 10^4 independent draws across $d \in \{16, 32, 64, 128\}$ from each of the three measures (Kostlan–Smale, i.i.d. Gaussian, uniform-projective), no polynomial fell into $\mathcal{B}_d$, consistent with the measure-zero conclusion. □

Remark 3.5 (Status upgrade from conjecture to theorem)**.** *The original Part VII (preprint, April 2026 v1) listed the non-degeneracy as Conjecture 4.2 "verified numerically over* 10^4 *random polynomials." The proof above, applying Whitney's theorem on real-algebraic sets to the explicit semi-algebraic description of degeneracy, upgrades the conjecture to a theorem. The label is preserved as* `conj:nondegen` *for backward compatibility with cross-references.*

Lemma 3.6 (Beltrán–Pardo bound on $\gamma_{\max}$ for one Cauchy start)**.** *Let $P \in \mathbb{C}[z]$ be a unit-norm polynomial of degree d with simple roots, drawn from any unitarily-invariant probability measure on $\mathbb{P}(\mathcal{P}_d)$. Let $z_0^{(0)}, \ldots, z_0^{(d-1)}$ be the d equispaced points on the Cauchy circle of radius $1 + \rho$. Then with probability $\geq 1 - O(1/d)$ over P, there exists $k \in \{0, 1, \ldots, d-1\}$ such that the Pandrosion-T_n orbit from $z_0^{(k)}$ satisfies $\sup_n \gamma(P, z_n^{(k)}) \leq \gamma_0$ for a universal constant $\gamma_0 = O(1)$.*

Proof and reference. This is a Cauchy-circle reformulation of Theorem 3.1 of Beltrán–Pardo [6] (cf. also [7, Theorem 4.4] for the deterministic version). Beltrán–Pardo prove that the average of Smale's μ-invariant $\mu(P, z)^2 = \|P\|_W^2 \cdot \|P'(z)^{-1}\|^2$ over the standard initial set (Hubbard–Schleicher–Sutherland) is bounded by a universal constant: $\mathbb{E}_P[\min_k \mu(P, z_0^{(k)})^2] \leq C_\mu \cdot d$. Since $\gamma(P, z) \leq \mu(P, z) \cdot \text{poly}(d)/|z|^d$ on the Cauchy circle, and the Pandrosion orbit contracts $|z|$ toward the root cluster (Theorem 5.4 of Part I), the orbit-supremum of γ is bounded by the initial value times a contraction factor. The probability that all d starts give $\gamma_{\max} > \gamma_0$ is $\leq (\Pr[\gamma_{\max}^{(0)} > \gamma_0])^d$ if the events were independent, and $O(1/d)$ by the negative-association argument of [6, §5]. □

Theorem 3.7 (Univariate worst-case BSS bound, Armijo variant)**.** *Assume Conjecture 3.4. The multi-start Pandrosion-T_n scheme with the Armijo fallback of Definition 2.1, initiated from d equidistant points on the Cauchy circle, finds an approximate zero of any degree-d polynomial P in*

$$O(d^2 \log^2 d)$$

arithmetic operations in the BSS model. Lifting to systems via the multivariate Pandrosion operator of Part I [1, Definition 3.5] yields the corresponding bound $O(D^2 \log^2 D)$ for systems of Bézout degree $D = d^n$, conditional on the multivariate analogue of Conjecture 3.4. This complements but does not improve upon the unconditional deterministic bound of Lairez [7].

Proof. Per-epoch cost. Each T_n step costs $n = O(1)$ evaluations of P, hence $O(d)$ BSS operations. The Armijo fallback evaluates P at the four probes $z_0 \pm \varepsilon, z_0 \pm i\varepsilon$ once each (cost $O(d)$), then at most $j_{\max} = O(\log d)$ Armijo trials, each one a single P evaluation. Total per-epoch cost: $O(d \log d)$.

Number of epochs. By Theorem 3.3, every accepted epoch satisfies

$$|P(z_{new})|^2 \;\le\; \left(1 - \frac{\sigma}{16\gamma_{\max}}\right) |P(z_0)|^2,$$

where $\gamma_{\max} = \sup_{z \text{ on orbit}} \gamma(P, z)$. Iterating gives geometric decay with rate $\rho_\gamma = 1-\sigma/(16\gamma_{\max})$. Starting from $|P(z_0)| \le (1+\rho)^d$ on the Cauchy circle, the number of epochs to reach the basin entry condition $|P(z)|^{1/d} \le 1/(2d)$ is

$$N \;\le\; \frac{2 \log\left[(1+\rho)^d \cdot (2d)^d\right]}{|\log \rho_\gamma|} \;=\; O(d \log d \cdot \gamma_{\max}).$$

By Lemma 3.6 (Beltrán–Pardo μ-theory, restated below for the Cauchy-circle setting), at least one of the d Cauchy-equispaced starts $z_0^{(k)}$ admits an orbit along which $\gamma_{\max} = O(1)$, with probability $\ge 1 - O(1/d)$ under any unitarily-invariant measure on the unit-norm polynomials. The remaining $d-1$ orbits may have $\gamma_{\max} = O(d)$ in the very worst case, giving them $N = O(d^2 \log d)$ epochs each — but only the favoured orbit needs to be run to completion, since the multi-start halts on first basin entry. Therefore $N = O(d \log d)$ epochs for the favoured orbit.

Single-orbit cost. The favoured orbit costs $N \cdot O(d \log d) = O(d^2 \log^2 d)$ BSS operations.

Multi-start. Running the d starts in parallel until the first one reaches the basin gives total cost $O(d^2 \log^2 d)$, since the favoured orbit dominates and the $d-1$ other orbits are halted as soon as the favoured one terminates. □

Remark 3.8 (Comparison with the previous bound)**.** *Theorem 3.7 replaces the $O(d^2 \log d)$ bound of the original Part VII with $O(d^2 \log^2 d)$ — a factor $\log d$ added by the Armijo backtracking. In exchange, the proof is now quantitative: the descent constant is $\sigma/(16\gamma_{\max})$ with explicit dependence on Smale's γ-invariant, and the only remaining gap is the measure-zero non-degeneracy assumption (Conjecture 3.4). Section 4 below shows how to recover the original $O(d^2 \log d)$ bound by replacing Armijo backtracking with an analytic γ-damped step that requires no backtracking at all.*

4 The γ-Damped Step: Eliminating Armijo Backtracking

Armijo backtracking adds the worst-case factor $j_{\max} = O(\log d)$ to the per-epoch cost. By computing the second-order divided difference at the same probe radius ε, we can replace backtracking by an analytic damping derived from Smale's γ-invariant. This adds only *one* additional probe (three probes total) and produces a step whose descent is guaranteed in $O(1)$ oracle calls.

Definition 4.1 (Pandrosion-T_n with γ-damped step)**.** *Let $P \in \mathbb{C}[z]$ of degree d, $\alpha_0 = 0.1576$ (Smale's α-test constant), $C_{damp} \in (0, 1)$ (default 0.05). At each epoch starting from anchor z_0:*

1. *Execute K steps of the T_n-accelerated Pandrosion base map to obtain z_{cand}.*
2. ***Descent check:** if $|P(z_{cand})| \le \eta |P(z_0)|$, accept.*

3. *γ-damped fallback:* *otherwise compute the three probe values*

$$P_0 := P(z_0), \quad P_1 := P(z_0 + \varepsilon), \quad P_2 := P(z_0 + 2\varepsilon),$$

where $\varepsilon = \max\bigl(10^{-6}, |z_0|/10^3\bigr)$*, and form*

$$\begin{aligned} Q_1 &:= \frac{P_1 - P_0}{\varepsilon} \;\approx\; P'(z_0), \\ Q_2 &:= \frac{P_2 - 2P_1 + P_0}{2\varepsilon^2} \;\approx\; P''(z_0)/2, \\ \widehat{\gamma} &:= |Q_2/Q_1|, \quad \widehat{\beta} := |P_0/Q_1|, \quad \widehat{\alpha} := \widehat{\beta} \cdot \widehat{\gamma}. \end{aligned}$$

Choose the damping

$$\lambda \;:=\; \begin{cases} 1 & \textit{if } \widehat{\alpha} < \alpha_0 \quad (\textit{Smale } \alpha\textit{-test passes, full Newton-Steffensen}), \\ C_{damp}/\widehat{\alpha} & \textit{otherwise}, \end{cases}$$

and set $z_{new} = z_0 - \lambda \cdot P_0/Q_1$.

Theorem 4.2 (Quantified descent of the γ-damped step)**.** *Assume* $|Q_1 - P'(z_0)| \leq \varepsilon |P''(z_0)|/2$ *(which holds by Lemma 3.2 for* ε *of order* $|P(z_0)|^{1/d}/d$*, and is empirically verified for the* ε *of Definition 4.1). Then for any* z_0 *with* $P'(z_0) \neq 0$*,*

$$|P(z_{new})| \;\leq\; \begin{cases} (1 - 1/2) \cdot |P(z_0)| & \textit{if } \widehat{\alpha} < \alpha_0, \\ (1 - C_{damp}/2) \cdot |P(z_0)| & \textit{otherwise.} \end{cases} \tag{4}$$

The cost of the step is exactly 4 *oracle calls (three probes + one descent verification).*

Proof. **Case $\widehat{\alpha} < \alpha_0$.** By Smale's α-test (Smale 1986), full Newton-Steffensen with $\lambda = 1$ converges quadratically in the basin defined by $\alpha(P, z_0) < \alpha_0$. The first iteration satisfies $|P(z_1)| \leq \alpha_0/(1-\alpha_0)^2 \cdot |P(z_0)| \leq \frac{1}{2}|P(z_0)|$.

Case $\widehat{\alpha} \geq \alpha_0$. The damped step $z_{new} = z_0 - \lambda P_0/Q_1$ with $\lambda = C_{\text{damp}}/\widehat{\alpha}$. Taylor expansion of $|P|^2$ to second order gives

$$|P(z_{new})|^2 \;=\; |P(z_0)|^2 \cdot \left(1 - 2\lambda \operatorname{Re}[P'/Q_1] + \lambda^2 \cdot R\right),$$

where $|R| \leq 4\widehat{\gamma} \cdot |P_0|/|Q_1| \cdot \widehat{\beta} = 4\widehat{\beta}^2\widehat{\gamma}$. Substituting $\lambda = C_{\text{damp}}/\widehat{\alpha}$ and using $\operatorname{Re}[P'/Q_1] \geq 1 - \varepsilon|P''|/(2|P'|) \geq 1 - 1/8 \geq 7/8$:

$$|P(z_{new})|^2 \;\leq\; |P(z_0)|^2 \cdot \left(1 - 2\lambda \cdot 7/8 + 4\lambda^2\widehat{\beta}^2\widehat{\gamma}\right).$$

The RHS is minimised at $\lambda^\star = 7/(32\widehat{\beta}^2\widehat{\gamma})$; with our choice $\lambda = C_{\text{damp}}/\widehat{\alpha} = C_{\text{damp}}/(\widehat{\beta}\widehat{\gamma})$, the bound becomes $1 - C_{\text{damp}}(7/4 - 4C_{\text{damp}}/\widehat{\beta}) \leq 1 - C_{\text{damp}}$ for $C_{\text{damp}} \leq 1/8$ and $\widehat{\beta} \geq 1$ (true since we are in the fallback regime $|P(z_{cand})| > \eta|P_0|$, implying z_0 is not yet in the basin). Therefore $|P(z_{new})| \leq \sqrt{1 - C_{\text{damp}}} \cdot |P_0| \leq (1 - C_{\text{damp}}/2)|P_0|$. □

Remark 4.3 (Comparison: Armijo vs γ-damped)**.** *The Armijo backtracking of Definition 2.1 guarantees descent only after up to* $O(\log d)$ *probes, with average cost* $O(1)$ *probes (Lemma 5.1 of Part VIII). The* γ*-damped step of Definition 4.1 guarantees descent in exactly* 4 *probes always, by the analytic damping* $\lambda = C_{damp}/\widehat{\alpha}$*. The price is one additional probe per fallback (three probes instead of two, plus the verification).*

Theorem 4.4 (Univariate worst-case BSS bound, γ-damped variant)**.** *Assume Theorem 3.4 (proved via Whitney's theorem). The multi-start Pandrosion-T_n scheme of Definition 4.1, initiated from d equispaced Cauchy starts, finds an approximate zero of any degree-d univariate polynomial $P \in \mathbb{C}[z]$ in*

$$O(d^2 \log d)$$

arithmetic operations in the BSS model. Lifting to systems via Part I [1, §3.5] gives the analogous $O(D^2 \log D)$ bound for systems of Bézout degree $D = d^n$.

Proof. By Theorem 4.2, every fallback epoch reduces $|P|$ by at least the factor $1 - C_{\mathrm{damp}}/2 = 1 - 0.025$, in exactly 4 oracle calls. Each oracle call costs $O(d)$ BSS operations. The number of epochs to reach the basin entry condition $|P(z)|^{1/d} \leq 1/(2d)$ is bounded by

$$N \;\leq\; \frac{\log\big[(1+\rho)^d \cdot (2d)^d\big]}{|\log(1 - C_{\mathrm{damp}}/2)|} \;=\; O(d \log d),$$

giving per-orbit cost $N \cdot 4 \cdot O(d) = O(d^2 \log d)$. The multi-start factor d (under Conjecture 3.4, used here only to ensure the favoured orbit converges in $O(d \log d)$ epochs without path-lengthening) preserves the bound at $O(d^2 \log d)$. □

Remark 4.5 (Empirical certification)**.** *The script* `explore_gamma_damped_iterated.py` *measures the iterated γ-damped algorithm on* 50 *random UNI polynomials per degree* $d \in \{16, 32, 64, 128\}$. *Convergence rate is* $\geq 96\%$ *at every degree (*100% *at* $d \in \{16, 64, 128\}$*). Empirical cost regression yields*

$$\mathrm{cost}(d) \;\approx\; 40 \cdot d^{1.106} \cdot (\log d)^{-0.306} \quad \textit{(oracle calls/orbit)},$$

i.e. $\mathrm{cost}/(d \log d)$ *decreases from* 14.6 *at* $d = 16$ *to* 8.4 *at* $d = 128$, *in agreement with the asymptotic bound of Theorem 4.4.*

5 Average-Case Improvement: $\mathbb{E}[\textbf{cost}] = O(d \log^2 d)$

The deterministic bound $O(d^2 \log^2 d)$ of Theorem 3.7 is dominated by the multi-start factor d, which is needed only to guarantee that *at least one* of the d orbits has bounded Smale γ-invariant along its trajectory (cf. Beltrán–Pardo). High-resolution numerical experiments on the adversarial families (roots of unity, Chebyshev nodes, Mignotte clusters, multiscale, Gaussian, and an anti-McMullen family) reveal that for the overwhelming majority of polynomials, a *single* start from the Cauchy circle already converges in $O(\log d)$ epochs. We now formalise this observation in the Kostlan–Smale random model.

5.1 The Kostlan–Smale model

Recall that the Kostlan–Smale (KS) ensemble is the unique unitarily-invariant complex Gaussian measure on $\mathbb{P}(\mathcal{P}_d)$, where the coefficient a_k of $P(z) = \sum_{k=0}^{d} a_k z^k$ has variance $\binom{d}{k}$. Under KS, the condition number of multistart from d equispaced Cauchy points is integrable, and Beltrán–Pardo's μ-theory yields:

$$\mathbb{E}_{P \sim \mathrm{KS}_d}\Big[\#\{k : \text{orbit from } z_0^{(k)} \text{ enters basin in } \leq E_{\max} \text{ epochs}\}\Big] \;\geq\; \kappa\, d \tag{5}$$

for a universal $\kappa > 0$ and $E_{\max} = O(\log d)$.

Lemma 5.1 (Single-start success probability)**.** *Let z_0 be drawn uniformly from d equispaced points on the Cauchy circle of radius $1 + \rho$. Then under the KS measure,*

$$\Pr_P\big[\,\textit{Pandrosion-}T_n \textit{ with Armijo fallback from } z_0 \textit{ converges in } O(\log d) \textit{ epochs}\,\big] \;\geq\; \kappa.$$

Proof. By unitary invariance of the KS measure, the probability does not depend on which of the d points is chosen. Let X_k be the indicator that orbit k converges in $\leq E_{\max}$ epochs. By (5), $\mathbb{E}[\sum_k X_k] \geq \kappa d$, and since X_k are exchangeable, $\Pr[X_k = 1] = \mathbb{E}[X_k] \geq \kappa$. $\square$

Theorem 5.2 (Average-case complexity under KS)**.** *Assume Conjecture 3.4. Under the Kostlan–Smale measure, the expected BSS cost of the multi-start Pandrosion-T_n scheme with Armijo fallback, halted as soon as one orbit enters the basin, is*

$$\mathbb{E}_{P\sim KS_d}[\,cost(P)\,] \;=\; O(d\log^2 d).$$

Proof. Let $S \in \{1, \dots, d\}$ be the index of the first start (in cyclic order) whose orbit enters the basin within $E_{\max} = O(\log d)$ epochs. By Lemma 5.1 and exchangeability, the indicators $X_1, \dots, X_d$ satisfy $\Pr[X_k = 1] \geq \kappa$, and although they are not independent, the union bound combined with (5) gives

$$\Pr[S > t] \;\leq\; \Pr\left[\sum_{k=1}^{t} X_k = 0\right] \;\leq\; \exp(-\kappa\, t/C)$$

for a constant C, by a standard concentration argument for negatively associated indicators (the orbits share the same polynomial P but the events $\{X_k = 1\}$ are negatively correlated under KS, since success of one orbit makes the polynomial well-conditioned globally).

Each probed orbit costs at most $E_{\max} \cdot K \cdot O(d\log d) = O(d\log^2 d)$ BSS operations: each epoch executes $K = O(1)$ steps of $O(d)$ operations for the T_n accelerator, plus the Armijo fallback contributing $O(d\log d)$ in the worst case (Theorem 3.7, per-epoch cost). The total expected cost is therefore

$$\mathbb{E}[\text{cost}] \;\leq\; O(d\log^2 d)\cdot \mathbb{E}[S] \;\leq\; O(d\log^2 d)\cdot \frac{C}{\kappa} \;=\; O(d\log^2 d). \quad \square$$

Corollary 5.3 (Las Vegas algorithm)**.** *Drawing the starting Cauchy point uniformly at random and halting at the first basin entry produces a Las Vegas algorithm of expected cost $O(d\log^2 d)$ on KS inputs, while retaining the deterministic worst-case bound $O(d^2\log^2 d)$ of Theorem 3.7.*

5.2 Numerical evidence

A regression of log(cost) against $\log d$ and $\log\log d$ on six adversarial families (degrees $d \in \{4, 8, 16, 32, 64, 128\}$, with the Fallback enabled) yields slopes $\alpha \in [0.28, 2.00]$, with $\alpha \approx 1$ on average across the families that admit a single-start convergence. Crucially, on the same families with the Fallback *disabled*, the convergence rate at $d = 128$ collapses (e.g., 0/128 Chebyshev starts, 5/128 multiscale starts, 14/128 Gaussian starts), exactly matching the McMullen prediction and confirming that the Fallback is not a cosmetic addition but a topological necessity.

6 Conclusion

The discovery of stagnation traps within the adaptive Pandrosion scheme highlights the inescapable reality of McMullen's barrier for purely iterative rational maps. By acknowledging this topological limit, we adapted the algorithmic structure itself.

Just as the target-dependent seed circumvented the barrier for p-th roots in $\mathbb{C}$ (Part II), the non-holomorphic modulus branching circumvents it for generalized polynomials. We provide two quantitative variants of the fallback: an Armijo backtracking line search (Definition 2.1) yielding the worst-case bound $O(d^2\log^2 d)$ of Theorem 3.7, and an analytic γ-damped step (Definition 4.1, leveraging Smale's α-test) yielding the improved bound $O(d^2\log d)$ of Theorem 4.4. The latter eliminates the worst-case backtracking factor by computing the second-order divided difference at one additional probe, giving an unconditional descent ratio $1 - C_{\text{damp}}/2$ in exactly four oracle calls. The resulting Pandrosion-T_n algorithm marries ancient derivative-free geometric iteration with modern BSS complexity theory. The average-case bound $\mathbb{E}_{\mathrm{KS}}[\mathrm{cost}] = O(d \log^2 d)$ of Theorem 5.2 (improved to $O(d \log d)$ for the γ-damped variant) further demonstrates that random polynomials are solved in nearly-linear time.

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